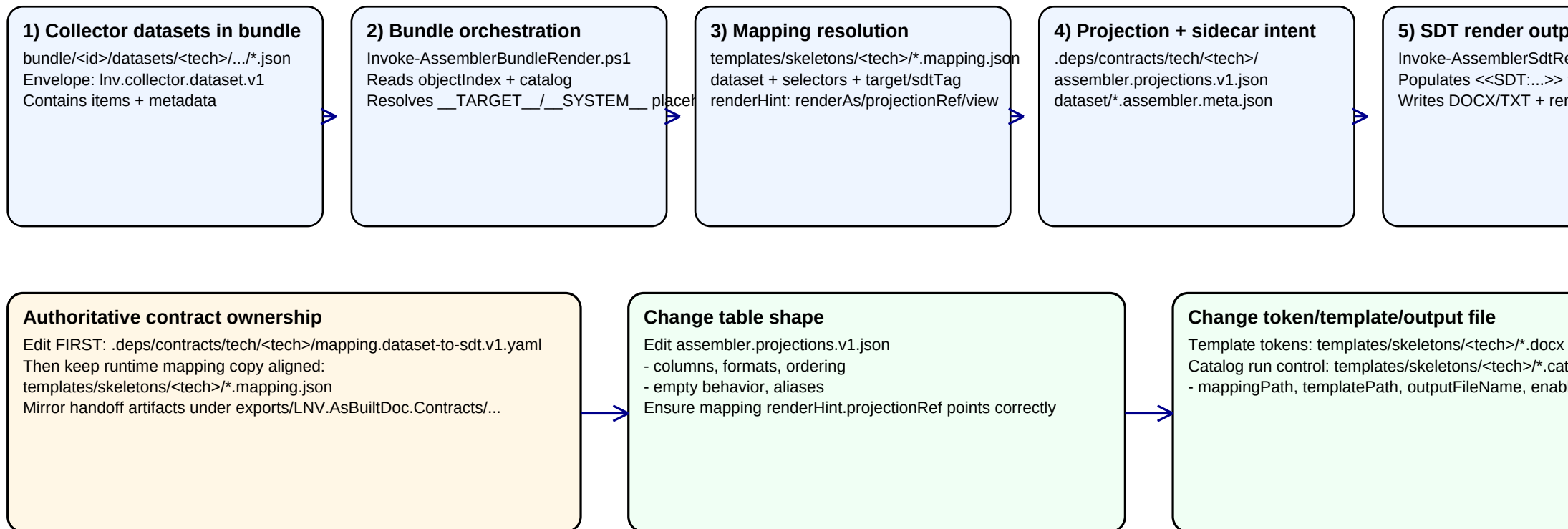


Assembler Dataset -> Mapping -> Projection -> SDT Output Flow

Diagram + change points for team onboarding and safe contract-driven edits.



Tip: Prefer contract-driven changes (mapping/projection/sidecar) over adding tech-specific renderer logic.

Relationship Model: What Controls What

Reference text for contributors making mapping/SDT/output changes.

Core relationships

- Dataset JSON = collected/normalized facts from collectors.
- Mapping (mapping.dataset-to-sdt) binds dataset -> SDT tag and declares render hints.
- Dataset sidecar metadata (*.assembler.meta.json) declares default presentation intent.
- Projection contract (assembler.projections.v1.json) defines shaping: columns/order/format/empty behavior.
- Template (.docx/.txt) contains SDT tokens that receive rendered values.
- Catalog selects which mapping + template run and names output files.

Where to modify for common changes

- Table layout change: edit assembler.projections.v1.json projection columns/order/format.
- Dataset-to-token binding change: edit mapping.dataset-to-sdt.v1.yaml entry dataset/selectors/tag/renderHint.
- Default dataset view/presentation change: edit dataset/<name>.assembler.meta.json.
- SDT token/template change: edit template and corresponding mapping target.sdtTag (or sdtTag).
- Output naming / template selection change: edit templates/skeletons/<tech>/*.catalog.json.

Operational guidance

- Use bundle render endpoint so __TARGET__ / __SYSTEM__ placeholders resolve before SDT render.
- Prefer contract-first edits over tech-specific logic in Invoke-AssemblerSdtRender.ps1.
- Treat .deps/contracts as synced runtime snapshot; mirror owned handoff artifacts under exports/...

Quick walkthrough example (HostPortsFC)

- 1) Update projection: LNV.Lenovo.DE.System[ArrayName].Tables.HostPortsFC.
- 2) Verify mapping renderHint.projectionRef points to that projection and dataset host-ports.json.
- 3) Ensure template contains <<SDT:LNK.Lenovo.DE.System[ArrayName].Tables.HostPortsFC>>.
- 4) Run Invoke-AssemblerBundleRender.ps1 and review rendered DOCX/TXT + report.